

14.2. Student's t -distribution

Here we treat the Student's t -distribution, which is also one of the very useful sampling distributions.

Definition 14.2. A continuous random variable X is said to have a t -distribution with ν degrees of freedom if its probability density function is of the form

$$f(x; \nu) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\pi\nu} \Gamma\left(\frac{\nu}{2}\right) \left(1 + \frac{x^2}{\nu}\right)^{\left(\frac{\nu+1}{2}\right)}, \quad -\infty < x < \infty$$

where $\nu > 0$. If X has a t -distribution with ν degrees of freedom, then we denote it by writing $X \sim t(\nu)$.

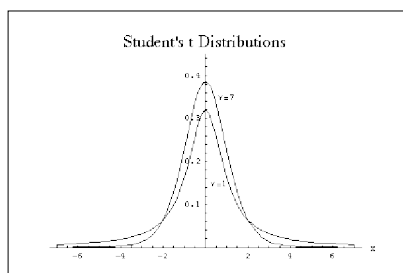
The t -distribution was discovered by W.S. Gosset (1876-1936) of England who published his work under the pseudonym of student. Therefore, this distribution is known as Student's t -distribution. This distribution is a generalization of the Cauchy distribution and the normal distribution. That is, if $\nu = 1$, then the probability density function of X becomes

$$f(x; 1) = \frac{1}{\pi(1+x^2)} \quad -\infty < x < \infty,$$

which is the Cauchy distribution. Further, if $\nu \rightarrow \infty$, then

$$\lim_{\nu \rightarrow \infty} f(x; \nu) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} \quad -\infty < x < \infty,$$

which is the probability density function of the standard normal distribution. The following figure shows the graph of t -distributions with various degrees of freedom.



Example 14.6. If $T \sim t(10)$, then what is the probability that T is at least 2.228 ?

Answer: The probability that T is at least 2.228 is given by

$$\begin{aligned} P(T \geq 2.228) &= 1 - P(T < 2.228) \\ &= 1 - 0.975 \quad (\text{from } t\text{-table}) \\ &= 0.025. \end{aligned}$$

Example 14.7. If $T \sim t(19)$, then what is the value of the constant c such that $P(|T| \leq c) = 0.95$?

Answer:

$$\begin{aligned} 0.95 &= P(|T| \leq c) \\ &= P(-c \leq T \leq c) \\ &= P(T \leq c) - 1 + P(T \leq c) \\ &= 2P(T \leq c) - 1. \end{aligned}$$

Hence

$$P(T \leq c) = 0.975.$$

Thus, using the t -table, we get for 19 degrees of freedom

$$c = 2.093.$$

Theorem 14.5. If the random variable X has a t -distribution with ν degrees of freedom, then

$$E[X] = \begin{cases} 0 & \text{if } \nu \geq 2 \\ DNE & \text{if } \nu = 1 \end{cases}$$

and

$$Var[X] = \begin{cases} \frac{\nu}{\nu-2} & \text{if } \nu \geq 3 \\ DNE & \text{if } \nu = 1, 2 \end{cases}$$

where DNE means does not exist.

Theorem 14.6. If $Z \sim N(0, 1)$ and $U \sim \chi^2(\nu)$ and in addition, Z and U are independent, then the random variable W defined by

$$W = \frac{Z}{\sqrt{\frac{U}{\nu}}}$$

has a t -distribution with ν degrees of freedom.

Theorem 14.7. If $X \sim N(\mu, \sigma^2)$ and $X_1, X_2, \dots, X_n$ be a random sample from the population X , then

$$\frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} \sim t(n-1).$$

Proof: Since each $X_i \sim N(\mu, \sigma^2)$,

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

Thus,

$$\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0, 1).$$

Further, from Theorem 14.3 we know that

$$(n-1) \frac{S^2}{\sigma^2} \sim \chi^2(n-1).$$

Hence

$$\frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} = \frac{\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}}{\sqrt{\frac{(n-1) S^2}{(n-1) \sigma^2}}} \sim t(n-1) \quad (\text{by Theorem 14.6}).$$

This completes the proof of the theorem.

Example 14.8. Let X_1, X_2, X_3, X_4 be a random sample of size 4 from a standard normal distribution. If the statistic W is given by

$$W = \frac{X_1 - X_2 + X_3}{\sqrt{X_1^2 + X_2^2 + X_3^2 + X_4^2}},$$

then what is the expected value of W ?

Answer: Since $X_i \sim N(0, 1)$, we get

$$X_1 - X_2 + X_3 \sim N(0, 3)$$

and

$$\frac{X_1 - X_2 + X_3}{\sqrt{3}} \sim N(0, 1).$$

Further, since $X_i \sim N(0, 1)$, we have

$$X_i^2 \sim \chi^2(1)$$

and hence

$$X_1^2 + X_2^2 + X_3^2 + X_4^2 \sim \chi^2(4)$$

Thus,

$$\frac{\frac{X_1 - X_2 + X_3}{\sqrt{3}}}{\sqrt{\frac{X_1^2 + X_2^2 + X_3^2 + X_4^2}{4}}} = \left(\frac{2}{\sqrt{3}} \right) W \sim t(4).$$

Now using the distribution of W , we find the expected value of W .

$$\begin{aligned} E[W] &= \left(\frac{\sqrt{3}}{2} \right) E \left[\frac{2}{\sqrt{3}} W \right] \\ &= \left(\frac{\sqrt{3}}{2} \right) E[t(4)] \\ &= \left(\frac{\sqrt{3}}{2} \right) 0 \\ &= 0. \end{aligned}$$

Example 14.9. If $X \sim N(0, 1)$ and X_1, X_2 is random sample of size two from the population X , then what is the 75th percentile of the statistic $W = \frac{X_1}{\sqrt{X_2^2}}$?

Answer: Since each $X_i \sim N(0, 1)$, we have

$$\begin{aligned} X_1 &\sim N(0, 1) \\ X_2^2 &\sim \chi^2(1). \end{aligned}$$

Hence

$$W = \frac{X_1}{\sqrt{X_2^2}} \sim t(1).$$

The 75th percentile a of W is then given by

$$0.75 = P(W \leq a)$$

Hence, from the t -table, we get

$$a = 1.0$$

Hence the 75th percentile of W is 1.0.

Example 14.10. Suppose $X_1, X_2, \dots, X_n$ is a random sample from a normal distribution with mean μ and variance σ^2 . If $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ and $V^2 =$

$\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$, and X_{n+1} is an additional observation, what is the value of m so that the statistics $\frac{m(\bar{X} - X_{n+1})}{V}$ has a t -distribution.

Answer: Since

$$\begin{aligned} X_i &\sim N(\mu, \sigma^2) \\ \Rightarrow \bar{X} &\sim N\left(\mu, \frac{\sigma^2}{n}\right) \\ \Rightarrow \bar{X} - X_{n+1} &\sim N\left(\mu - \mu, \frac{\sigma^2}{n} + \sigma^2\right) \\ \Rightarrow \bar{X} - X_{n+1} &\sim N\left(0, \left(\frac{n+1}{n}\right) \sigma^2\right) \\ \Rightarrow \frac{\bar{X} - X_{n+1}}{\sigma \sqrt{\frac{n+1}{n}}} &\sim N(0, 1) \end{aligned}$$

Now, we establish a relationship between V^2 and S^2 . We know that

$$\begin{aligned} (n-1)S^2 &= (n-1) \frac{1}{(n-1)} \sum_{i=1}^n (X_i - \bar{X})^2 \\ &= \sum_{i=1}^n (X_i - \bar{X})^2 \\ &= n \left(\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \right) \\ &= nV^2. \end{aligned}$$

Hence, by Theorem 14.3

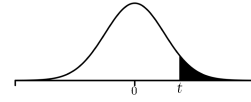
$$\frac{nV^2}{\sigma^2} = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1).$$

Thus

$$\left(\sqrt{\frac{n-1}{n+1}} \right) \frac{\bar{X} - X_{n+1}}{V} = \frac{\frac{\bar{X} - X_{n+1}}{\sigma \sqrt{\frac{n+1}{n}}}}{\sqrt{\frac{\frac{nV^2}{\sigma^2}}{(n-1)}}} \sim t(n-1).$$

Thus by comparison, we get

$$m = \sqrt{\frac{n-1}{n+1}}.$$

Critical Values for Student's t -Distribution.

df	Upper Tail Probability: $\Pr(T > t)$									
	0.2	0.1	0.05	0.04	0.03	0.025	0.02	0.01	0.005	0.0005
1	1.376	3.078	6.314	7.916	10.579	12.706	15.895	31.821	63.657	636.619
2	1.061	1.886	2.920	3.320	3.896	4.303	4.849	6.965	9.925	31.599
3	0.978	1.638	2.353	2.605	2.951	3.182	3.482	4.541	5.841	12.924
4	0.941	1.533	2.132	2.333	2.601	2.776	2.999	3.747	4.604	8.610
5	0.920	1.476	2.015	2.191	2.422	2.571	2.757	3.365	4.032	6.869
6	0.906	1.440	1.943	2.104	2.313	2.447	2.612	3.143	3.707	5.959
7	0.896	1.415	1.895	2.046	2.241	2.365	2.517	2.998	3.499	5.408
8	0.889	1.397	1.860	2.004	2.189	2.306	2.449	2.896	3.355	5.041
9	0.883	1.383	1.833	1.973	2.150	2.262	2.398	2.821	3.250	4.781
10	0.879	1.372	1.812	1.948	2.120	2.228	2.359	2.764	3.169	4.587
11	0.876	1.363	1.796	1.928	2.096	2.201	2.328	2.718	3.106	4.437
12	0.873	1.356	1.782	1.912	2.076	2.179	2.303	2.681	3.055	4.318
13	0.870	1.350	1.771	1.899	2.060	2.160	2.282	2.650	3.012	4.221
14	0.868	1.345	1.761	1.887	2.046	2.145	2.264	2.624	2.977	4.140
15	0.866	1.341	1.753	1.878	2.034	2.131	2.249	2.602	2.947	4.073
16	0.865	1.337	1.746	1.869	2.024	2.120	2.235	2.583	2.921	4.015
17	0.863	1.333	1.740	1.862	2.015	2.110	2.224	2.567	2.898	3.965
18	0.862	1.330	1.734	1.855	2.007	2.101	2.214	2.552	2.878	3.922
19	0.861	1.328	1.729	1.850	2.000	2.093	2.205	2.539	2.861	3.883
20	0.860	1.325	1.725	1.844	1.994	2.086	2.197	2.528	2.845	3.850
21	0.859	1.323	1.721	1.840	1.988	2.080	2.189	2.518	2.831	3.819
22	0.858	1.321	1.717	1.835	1.983	2.074	2.183	2.508	2.819	3.792
23	0.858	1.319	1.714	1.832	1.978	2.069	2.177	2.500	2.807	3.768
24	0.857	1.318	1.711	1.828	1.974	2.064	2.172	2.492	2.797	3.745
25	0.856	1.316	1.708	1.825	1.970	2.060	2.167	2.485	2.787	3.725
26	0.856	1.315	1.706	1.822	1.967	2.056	2.162	2.479	2.779	3.707
27	0.855	1.314	1.703	1.819	1.963	2.052	2.158	2.473	2.771	3.690
28	0.855	1.313	1.701	1.817	1.960	2.048	2.154	2.467	2.763	3.674
29	0.854	1.311	1.699	1.814	1.957	2.045	2.150	2.462	2.756	3.659
30	0.854	1.310	1.697	1.812	1.955	2.042	2.147	2.457	2.750	3.646
31	0.853	1.309	1.696	1.810	1.952	2.040	2.144	2.453	2.744	3.633
32	0.853	1.309	1.694	1.808	1.950	2.037	2.141	2.449	2.738	3.622
33	0.853	1.308	1.692	1.806	1.948	2.035	2.138	2.445	2.733	3.611
34	0.852	1.307	1.691	1.805	1.946	2.032	2.136	2.441	2.728	3.601
35	0.852	1.306	1.690	1.803	1.944	2.030	2.133	2.438	2.724	3.591
36	0.852	1.306	1.688	1.802	1.942	2.028	2.131	2.434	2.719	3.582
37	0.851	1.305	1.687	1.800	1.940	2.026	2.129	2.431	2.715	3.574
38	0.851	1.304	1.686	1.799	1.939	2.024	2.127	2.429	2.712	3.566
39	0.851	1.304	1.685	1.798	1.937	2.023	2.125	2.426	2.708	3.558
40	0.851	1.303	1.684	1.796	1.936	2.021	2.123	2.423	2.704	3.551
41	0.850	1.303	1.683	1.795	1.934	2.020	2.121	2.421	2.701	3.544
42	0.850	1.302	1.682	1.794	1.933	2.018	2.120	2.418	2.698	3.538
43	0.850	1.302	1.681	1.793	1.932	2.017	2.118	2.416	2.695	3.532
44	0.850	1.301	1.680	1.792	1.931	2.015	2.116	2.414	2.692	3.526
45	0.850	1.301	1.679	1.791	1.929	2.014	2.115	2.412	2.690	3.520
46	0.850	1.300	1.679	1.790	1.928	2.013	2.114	2.410	2.687	3.515
47	0.849	1.300	1.678	1.789	1.927	2.012	2.112	2.408	2.685	3.510
48	0.849	1.299	1.677	1.789	1.926	2.011	2.111	2.407	2.682	3.505
49	0.849	1.299	1.677	1.788	1.925	2.010	2.110	2.405	2.680	3.500
50	0.849	1.299	1.676	1.787	1.924	2.009	2.109	2.403	2.678	3.496
60	0.848	1.296	1.671	1.781	1.917	2.000	2.099	2.390	2.660	3.460
70	0.847	1.294	1.667	1.776	1.912	1.994	2.093	2.381	2.648	3.435
80	0.846	1.292	1.664	1.773	1.908	1.990	2.088	2.374	2.639	3.416
90	0.846	1.291	1.662	1.771	1.905	1.987	2.084	2.368	2.632	3.402
100	0.845	1.290	1.660	1.769	1.902	1.984	2.081	2.364	2.626	3.390
120	0.845	1.289	1.658	1.766	1.899	1.980	2.076	2.358	2.617	3.373
140	0.844	1.288	1.656	1.763	1.896	1.977	2.073	2.353	2.611	3.361
180	0.844	1.286	1.653	1.761	1.893	1.973	2.069	2.347	2.603	3.345
200	0.843	1.286	1.653	1.760	1.892	1.972	2.067	2.345	2.601	3.340
500	0.842	1.283	1.648	1.754	1.885	1.965	2.059	2.334	2.586	3.310
1000	0.842	1.282	1.646	1.752	1.883	1.962	2.056	2.330	2.581	3.300
∞	0.842	1.282	1.645	1.751	1.881	1.960	2.054	2.326	2.576	3.291
	60%	80%	90%	92%	94%	95%	96%	98%	99%	99.9%
	Confidence Level									

Note: $t(\infty)_{\alpha/2} = Z_{\alpha/2}$ in our notation.